

Worksheet II

Introduction to Functions

Name: _____

Instructions

Use **pen**. Write questions and surprises in the margin in a different color. Some things in the reading are intentionally imprecise — find them.

Part 1 — What Is a Function?

[45 pts]

Reading

Read this section before attempting any problems.

The Definition

A **function** f from set A to set B , written $f: A \rightarrow B$, is a rule that assigns to each element of A exactly one element of B .

Think of a vending machine: you enter a code (input) and always receive exactly one item (output). A machine that sometimes gives you nothing, or sometimes gives two items for the same code, is broken. A function is a vending machine that is never broken.

Three things to name:

Term	Meaning
Domain (A)	The set of allowed inputs.
Codomain (B)	The set of declared outputs.
Range	The set of outputs actually produced.

The range is always a subset of the codomain, but they are not always equal. Most textbooks blur this distinction. We will not.

Notation

$f(x)$ is read “ f of x ” — the output of f when x is the input. It is **not** f times x .

Example. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$.

$$\begin{aligned}f(3) &= 9 \\f(-2) &= 4 \\f(0) &= 0\end{aligned}$$

Domain: $\mathbb{R}$. Codomain: $\mathbb{R}$. Range: all positive real numbers.

Injective Functions

A function f is **injective** (also called one-to-one) if different inputs always produce different outputs. Formally:

$$f(a) = f(b) \implies a = b$$

Is $f(x) = x^2$ injective? No. $f(2) = 4$ and $f(-2) = 4$, but $2 \neq -2$. Two different inputs produced the same output.

Is $f(x) = 2x + 1$ injective? Suppose $f(a) = f(b)$. Then $2a + 1 = 2b + 1$, so $2a = 2b$, therefore $a = b$. **Yes.**

Notice the proof structure: *suppose* the outputs are equal, derive that the inputs must be equal. This is your template.

Surjective Functions

A function $f: A \rightarrow B$ is **surjective** (also called onto) if every element of B is hit by at least one input. Formally:

$$\text{for every } b \in B, \text{ there exists } a \in A \text{ such that } f(a) = b$$

Is $f(x) = x^2$ surjective as $f: \mathbb{R} \rightarrow \mathbb{R}$? No: -1 is in the codomain but $f(x) = -1$ has no real solution.

What if we redefine the codomain as $[0, \infty)$? Then range equals codomain, so yes — surjective. Surjectivity depends on the codomain you declare.

Bijjective Functions

A function is **bijjective** if it is both injective and surjective. Every element of the codomain is hit by exactly one input — a perfect pairing. Only bijjective functions have inverses.

Foundation Practice

1. [15 pts] Let $f(x) = 3x - 2$. Fill in the table, then answer the questions below.

x	$f(x)$
0	
1	
-1	
4	
$\frac{1}{3}$	

- (a) What is $f(f(1))$? Answer: _____
- (b) Find x such that $f(x) = 0$. Answer: _____
- (c) Is f injective? Write a one-sentence proof using *suppose ... then ... therefore*.

2. [15 pts] Decide whether each rule defines a valid function from $A = \{1, 2, 3\}$ to $B = \{1, 2, 3, 4, 5\}$. If it does, state whether it is injective and/or surjective. If it does not, explain why not.

- (a) Each number maps to its square: $1 \mapsto 1$, $2 \mapsto 4$, $3 \mapsto 9$.

(b) Every number maps to 1: $1 \mapsto 1$, $2 \mapsto 1$, $3 \mapsto 1$.

(c) $1 \mapsto 2$, $2 \mapsto 4$, $2 \mapsto 5$.

3. [15 pts] For each function, state the domain, codomain, and range.

(a) $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^2 + 1$.

(b) $h: \{1, 2, 3\} \rightarrow \{1, 4, 9\}$ defined by $h(x) = x^2$.

Is h surjective? Compare your answer to (a) and explain why they differ despite using the same formula.

Part 2 — Application

[35 pts]

4. [20 pts] From Bedrooms to Functions.

In Homework 1 Part 4, you were carpeting two Minecraft bedrooms. Both rooms are 10 blocks long. Bedroom 2 is 8 blocks wide. Bedroom 1 has unknown width w . Each carpet block takes 2 wool.

- (a) Write a function $f(w)$ giving the **total** wool needed for both rooms.

- (b) Evaluate $f(6)$. What does this number mean in context?

- (c) Find w such that $f(w) = 300$. What does this answer mean in context?

- (d) Is f injective? Write a formal proof using *suppose ... then ... therefore*.

(e) State the natural domain of f in the Minecraft context. Is f surjective onto the positive integers? Explain.

5. [15 pts] Speaking Functions.

(a) Read $f(x) = 3x^2 + 1$ as a complete English sentence. Write what you would say.

(b) A classmate says: “ $f(x)$ means f times x .” Are they right? Correct them in one sentence.

(c) Write $f(g(x))$ as a complete English sentence. What do you think this expression means?

Part 3 — Challenge

[20 pts]

6. [8 pts] The Finite Trap.

Try to build a function $f: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ that is injective but *not* surjective. Spend two minutes trying. Then explain what happens and why.

7. [12 pts] The Collatz Function.

Define $C: \mathbb{N} \rightarrow \mathbb{N}$ by:

$$C(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases}$$

Starting from any positive integer, apply C repeatedly.

- (a) Compute the sequence starting at $n = 6$. Stop when you reach 1.

- (b) Compute the first 15 terms starting at $n = 27$.

(c) Is C injective? Either prove it or find a counterexample.

(d) Write a conjecture about what always happens when you apply C repeatedly starting from any positive integer.

Part 4 — Notebook Artifact

(ungraded)

On the **left page** of your notebook, write:

1. Two **Question Compass** questions about the word *function*. Pick any two directions.
2. One **muddiest point** — the single thing you are least sure about after today.